Phase 3J | First structural assembly

Pulsipher Farms / Moapa Valley Corn Maze

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This revision joins the Phase 3I member alternatives into a single J1a coordination model, calculates transparent whole-frame load paths, and defines a representative seat/floor prototype and first-article records. It advances the design for engineer and fabricator review; it does not release steel cutting, welding, passenger testing or operation.

Decision	Phase 3J position
Frame study	Carry the 16 deeper central crossmembers and 52 wider seat rails into J1a for coordinated evaluation. Preserve all 110 member IDs and end planes.
Bolster roots	Retain baseline central and outer bolster geometry. Do not adopt the Phase 3I trial angles or tapered boxes as accepted connections; actual support geometry and stiffness remain decisive.
Seat and floor	Keep the final I-SH1 hardpoint as a separate four-foot prototype study. Do not silently add it to the J1a frame mass or imply supplier-qualified seating.
What can proceed	Compare exact geometry, identify joint demands, reserve equipment space, prepare material/supplier return forms, arrange the engineer-led prototype plan and define measured records.
Release conditions	Accepted design loads and local/fatigue analysis; exact bogie/seat/panel interfaces; released drawings, material/toughness, weld procedure and inspection criteria.

Retained owner directions: diesel rail train; 36-inch gauge under evaluation; three regular plus one accessible coach; zero-percent longitudinal grade target; air service brakes with spring-applied, air-released parking; North Pole location deferred. The public edition contains technical information only.

01 | Geometry and mass reconciliation

Item	Dimension or change	Status
Frame envelope	400 × 110 in; x -47 to 353, y ±55	Study
Main rails	Centers y ±34; HSS 12 × 4 × 1/4	Candidate
Bogie support stations	x45 and x261; 216 in spacing	Interface reference
Floor	32 in above rail in F1/E1/J1a; steel top31.003937in	Integration study
Central crossmembers	16 × HSS6 × 2 × 3/16; 64 in lengths	I-CM6 incorporated
Seat rails	52 × HSS3 × 2 × 3/16; 21/22 in lengths	I-S32 incorporated
J1a catalog member mass	1,816.347143 kg	110 members; excludes connections
Prior I combined comparison	1,938.154087 kg	Different scope: includes hardpoints and bolster alternatives

J1a = 1,527.287845 kg baseline + 180.759583 kg deeper crossmembers + 108.299714 kg wider seat rails. Its increase is 289.059298 kg. The Phase 3I combined comparison also included 121.537676 kg of seat hardpoints and a 0.269268 kg bolster alternative delta. These are not included in J1a. Neither subtotal is a completed coach tare.

Catalog mass and nominal sharp-corner CAD volume are separate quantities. Final fabrication will require actual tube corner radii, wall tolerance, measured supply lengths, weld metal, fasteners, coatings, equipment and weighed completed mass. A deeper crossmember reduces underfloor clearance by two inches; the complete bogie/brake envelope remains unverified.

The 13,200 kg upper regular-coach case remains an analytical load envelope, not a rating. Do not add the J1a steel subtotal on top of it. Any movement of mass into this frame must be reconciled inside the complete tare and payload ledger.

02 | Whole-frame load paths

Preliminary analytical study - not released for fabrication. Retains Phase 3I alternatives as separate options. The J1a member-only analytical set includes CM6/S32 changes; I-BA, I-BT and I-SH1 geometry are not selected into that set. This phase adds a reproducible whole-coach beam model; it does not install a selected joint or certify the frame.

Primary section-property source: J-L01: Atlas Tube, ASTM A500 rectangular HSS section properties, page2, HSS12×4×¼ row. Accessed October18,2026 (America/Los_Angeles). https://www.atlastube.com/wp-content/uploads/2018/04/Atlas-HSS-Section-Properties-A500-Rectangles.pdf

What the study establishes

The retained two idealized main rails develop about 54.54 kN·m peak vertical bending each and 19.97 mm elastic displacement in the balanced 13,200 kg, 2g sensitivity. Moving 600 kg from distributed tare to the front platform raises the front support demand and nearly doubles front support hogging, even though centre-span sagging decreases. Equipment placement therefore needs a controlled mass/CG schedule before selecting joints.

A single centre support must transmit a roll couple when the two main-rail reactions differ. A freely rolling point bearing cannot by itself balance that case. Paired support probes reduce the transverse beam lever arm, but their position, preload, contact law, articulation and wheel-load effects still require the actual bogie design.

Geometry, assumptions and coordinate convention

Inches: x increases from front interior wall toward rear; y is transverse; z is top of rail. Main beams run x=−47 to353, with vertical supports at45 and261: 216 in between supports and92 in overhang at each end. Main centres are y=±34. The constant section is retained HSS12×4×¼, strong I=119in⁴/S=19.8in³; weak I=21.0in⁴/S=10.5in³, design wall0.233in. These are verified Atlas catalog properties, not certified material or approved section selection.

Two continuous Euler-Bernoulli beams are solved by singularity-function integration, with zero displacement at both supports and free outer ends. Exact polynomial stationary roots locate bending and displacement peaks. Applied vertical forces are positive upward, gravity negative; positive bending is sagging and positive displacement upward. No body-shell composite stiffness is claimed. Support/frame flexibility and shear deformation are excluded.

Mass elements are projected onto y=±34 by static transverse resultant balance. This is a load distribution assumption, not a solved flexible grillage. A lateral acceleration probe adds roll transfer using a 32 in CG-to-effective-support lever; that lever is unmeasured. Lateral main-rail bending uses a separate equal 50/50 horizontal-force split. Real torsion, diaphragms and attachment stiffness may change these distributions.

Reconciled mass - no double counting

Item	kg	Status
Regular tare allocation	7,040.000	Retained budget; not measured
J1a member-only steel	1,816.347	CM6/S32 analytical set; excludes I-BA/I-BT/I-SH1 geometry
I-SH1 analytical mass allowance	121.538	Distributed at64 feet; not installed in J1a geometry
16 bench furniture allowances	240.000	Contained within tare; 15kg each
Remaining unresolved tare	4,862.115	Body/equipment/bogies and unlisted hardware
32 occupants at145kg	4,640.000	Load sensitivity
One crew at145kg	145.000	Local study allowance
Unallocated upper-case reserve	1,375.000	Brings total to proposed upper case
Total upper study mass	13,200.000	Not rated occupancy or weight

The 261 input mass elements preserve the110 retained member masses and the68 CM6/S32 mass changes, 64 hardpoint mass allowances,16 occupied305kg benches, residual tare, reserve and crew. Bench loads use their gravity resultant at x41,73,...265 and y±32; local four-foot forces remain in Phase3I. This is valid for global force/first-moment balance, not local seat-rail bending. Residual tare and upper reserve are initially uniform over the312 in cabin. The 600 kg fore/aft test removes exactly600 kg from residual uniform tare and reapplies it at x−25 or331. It is an equipment-placement sensitivity, not a real specified component.

All 13,200 kg is applied to the body frame because actual unsprung/bogie masses are unknown. This overstates total body support weight relative to subtracting unsprung mass; it is not a universal upper bound on local stresses or dynamic response.

Main-rail demand digest

The following values refer to the y=+34 rail; negative lateral cases mirror the rail sides. The 600 kg rear case mirrors front longitudinally.45 combinations cover three distributions, three vertical factors and lateral accelerations0,±0.1g,±0.2g. None is an adopted regulatory load case.

Case	Front / rear reaction kN	Max vertical M kN·m	Gross vertical MPa	Max elastic deflection mm
Balanced; 1g vertical	32.362 / 32.362	27.269	84.04	9.985
Balanced; 1.5g vertical	48.543 / 48.543	40.904	126.07	14.977
Balanced; 2g vertical	64.724 / 64.724	54.538	168.09	19.969
600 kg moved to front; 2g	69.573 / 59.875	47.153	145.33	16.911
Balanced; 1g + 0.2g lateral	38.454 / 38.454	32.402	99.86	11.864
600 kg front; 2g + 0.2g lateral	76.121 / 65.510	51.591	159.00	18.502

Balanced 2g support bending is−10.702kN·m at both x45/261. The front600 kg2g case is−20.612/−10.150kN·m. These are signed main-beam section moments at support stations, not an approved connection demand envelope. The largest separate-maxima vertical-plus-lateral corner-stress bound among all cases is 215.604MPa. Adding separate maxima is conservative for the assumed two beam models; it does not include torsion, axial coupler loads or local stress concentration. No allowable stress, deflection limit or fatigue acceptance has been established.

Bolster support sensitivity

Each bogie is given the reactions from its two main rails, applied downward at y±34. The transverse member comparison uses a continuous gross 12 × 4 × 1/4 section, including the main-centre intersection region as an analytical idealization. The physical central member ends at y±32; these two inches each side are not a fabricated extension. Their stiffness/load introduction must be supplied by the joint. This study does not assign root joint stiffness or select I-BA.

For the balanced 2g case each bogie receives129.448kN. A centre point support gives55.896kN·m peak transverse bending (172.271MPa gross M/S). Equal paired supports give:

Paired support probe	Max gross bending MPa	Max elastic displacement mm
y=±18in	81.068	0.6395
y=±24in	50.668	0.3084
y=±30in	20.267	0.1927
y=±34in	0.000	0.0000

The y ±34 case places ideal supports directly under the applied loads and therefore removes ideal transverse span bending. It says nothing about local bearing, mounting space, welds, fatigue or whether a bogie can be built that way; it is not a selected geometry. Paired supports are unilateral unless positive retention is engineered. No uplift occurs in the listed 45 probes, but that does not establish derailment, crosslevel or wheel-unloading performance.

Balanced 1g plus0.2g lateral produces26.270/38.454kN main reactions at each bogie and requires a10.522kN·m centre-support roll couple. The paired y±18 reactions become20.855/43.868kN. A centre bearing with no roll restraint is incompatible with that nonzero couple; a real centre-bearing/side-bearer system needs a defined contact, preload and roll-stop law.

Signed moments evaluated at y±32 are section cuts inside this idealized beam, not resolved joint end moments. They must not be copied into I-BA weld sizing as a favorable new boundary condition. Moment sharing among the main rails, bolster, side bearings and body requires a stiffness model using the actual interfaces.

Verification and reproducibility

114 checks pass: closed-form simply supported UDL and centre-point-load reactions, bending and displacement; total 13,200 kg mass balance; all beam force and moment equilibrium; prescribed support displacements; and fore/aft mirror symmetry. This verifies the analytical implementation within its assumptions. It is not independent structural design approval.

Run loadpath_calc.py and check_loadpath.py in this folder. loadpath_inputs.json is the frozen, portable input, including source hashes and261 mass entries. build_loadpath_inputs.py is the provenance regeneration script and additionally needs the preserved phase3i sibling source; the calculation itself has no predecessor-file dependency. Full results are in loadpath_results.json; drawing_hooks.json contains the six digest rows and balanced 2g bolster data.

Required decisions before fabrication selection

1. Obtain actual bogie centre-bearing and side-bearer positions, footprints, stiffness/contact law, retention, articulation and force envelope. Include wheel-load and track-twist analysis.

2. Place and weigh real body, HVAC, batteries/air equipment, glazing, stairs, panels, bogies and seats. Separate sprung/unsprung mass; approve CG and equipment-placement limits.

3. Include coupler tension/compression, braking loads, longitudinal/lateral attachment forces, twist, impact and fatigue spectrum in one coordinated design basis.

4. Solve main/bolster joint rotational stiffness, local HSS wall load introduction and body racking. Check global stability, torsion, local buckling and cumulative fatigue. No generic plate or weld is selected here.

5. Agree deformation limits for doors, floor, glazing, couplings and boarding, plus weld/material/toughness and prototype acceptance criteria.

03 | First prototype and engineering returns

First assembly: a complete seat and floor bay

Develop one complete seat/floor bay before committing the entire coach frame to shop fabrication. Carry I-CM6, I-S32 and the revised 3/8-inch-cap I-SH1 into that prototype design. Preserve the F1/H1 source frame. Do not promote the I-BA twin-angle bolster root to the integrated structural candidate: its larger root-moment cases exceeded the limited weld comparison and its largest gross angle stress exceeded the candidate A36 yield comparison. Actual bogie support geometry and a compatible local/global load path must drive that redesign.

This is a decision about the next engineering article, not a production release. An empty, nonstructural full-size layout can establish fit and service access now. Structural specimens require a separate accepted joint, material, fixture and test plan. No people are used as ballast in development testing.

Family	Phase 3J development disposition	Required evidence
Central crossmembers	I-CM6, HSS 6×2×3/16, in pilot	Receiving wall, effective weld length, joint stiffness/fatigue and clearance below the deeper member
Seat rails and mounts	I-S32, HSS 3×2×3/16, with I-SH1 steel bypass	Actual seat-to-sole joint, bolt assembly, outer receiver and combined-load behavior
Central bolster root	I-BA retained only as comparison; excluded from integrated candidate	OEM support/side-bearer law, local load spreading and accepted root connection
Outer bolster	I-BT geometric option retained	Body reactions, perimeter reinforcement, seam/end-joint design and corrosion access
Floor	I-FB/I-FR separable bearing/retention demonstrator	Actual panel build, closeouts/inserts, clip spacing, moisture and fire performance

The earlier all-options mass comparison of 1,938.154 kg includes the unselected angle/taper combination and is not the mass of this development selection or the coach tare. Count the wider seat-rail mass change only once.

Pilot dimensions and connections

J-PILOT-01 is the exact source bay **x92-118, y−55-55 inches**, a **26×110-inch digital/empty fit-up envelope**. Crossmember centres are x93 and 117, 24 inches apart. Main girder centres remain y±34. Steel top is z31.003937 inches and the floor study is z32 inches. Two central crossmembers span y−32 to32; four outer receiver segments retain the thinner HSS 4×2×1/8. Four I-S32 seat rails occupy x94-116 at y−44,−20,20,44, each 22 inches between end planes.

One complete right bench, HB03R, has feet at **(97,20), (97,44), (113,20), (113,44)**. Its foot pattern is 16 inches longitudinally by 24 inches transversely. Each foot has two bolt axes at ±2 inches transversely from its centre. This arrangement exercises both a central CM6 receiver and a thin outer receiver. It does not substitute the stronger central joint for every seat connection.

The 26-inch crop is not a structural specimen cut length. Main and perimeter members are cropped visual context. The test engineer must provide fixture tails, load-development lengths, reaction restraints and an accepted fixture drawing. A short rigidly clamped coupon can conceal the receiving-wall flexibility that matters in the complete frame. Validated HSS analysis needs realistic boundaries and geometry; a cropped CAD model does not provide that validation. Steel Tube Institute

Per-foot component	Quantity for one bench	Defined study geometry	Held production detail
Cap	4	2.50×6.00×0.375 in, candidate A36	Actual plate properties, tolerances and joint acceptance
Side cheek	8	2.50×2.00×0.1875 in, candidate A36	Actual HSS corner fit and weld procedure
Soleplate	4	4.25×6.50×0.250 in, candidate A36	Actual seat-to-sole attachment
Steel spacer	8	1.000 in OD,0.5625 in bore,15.775 mm nominal length	Measured stack including coatings, flatness and isolation
Bolt assembly	8	Candidate 1/2-13 SAE Grade 5;9/16 in plate holes on4 in pitch	Length, washers, nut, locking and tightening

The seat load path is seat→soleplate→rigid spacers→cap/cheeks→seat rail→receiver→frame. The sandwich floor does not carry the seat-bolt clamping force. Four finished panel openings are 2.75×6.25 inches in the pilot; their OEM closeout and seal details remain unassigned. The soleplate is1/4 inch above the study floor, so the real seat must cover/protect the edge and preserve the intended seat height.

The original trial weld geometries are explicit: CM6 endpoints use two 3/16-inch legs on 4.5-inch drawn paths; seat-rail endpoints use two 3/16-inch legs on 2-inch paths. Each hardpoint has four 1/8-inch by 1.5-inch vertical cheek paths and two 3/16-inch by 2.5-inch cap paths. These dimensions identify test-development details, not approved welds. Actual root gaps, effective lengths, preparation, filler, WPS, qualifications, fatigue and inspection acceptance remain to be assigned.

The selected pilot is not the tightest access location. Also inspect an x49 end-of-rail hardpoint: the nominal 0.75-inch cap-to-crossmember gap becomes 0.5625 inch after the trial end-fillet projection. Test the actual socket, nut/washer, welding torch and inspection sight line. A nominal tool cylinder is not a demonstrated maintenance procedure.

Supplier return package

The editable JSON provides **38 blank technical-return fields across four forms**. Nothing has been sent. Each return must identify supplier, exact model, drawing revision, units, conditions, supporting evidence and reviewer disposition.

Form	Required return	What it resolves
J-IF-BG,12 fields	Rated load cases; wheel/rail geometry; complete mount footprint; bearing/side-bearer gaps and stiffness; yaw/roll/pitch; retention; brake reactions	Bogie support boundary and local frame redesign
J-IF-BR,10 fields	Exact chamber; service/spring force-versus-stroke; bracket/pushrod; rigging; pressure/control/failure matrix; hose motion; manual and rail-use acceptance	Air brake mechanical and pneumatic integration
J-IF-SE,8 fields	Seat geometry/mass/CG; actual foot coordinates and attachment; load cases; reactions; hardware; covers and durability	The missing seat-to-sole connection and accepted seat load path
J-IF-PN,8 fields	Panel construction; openings; bearing limits; rated inserts; closeouts; tolerances/environment; installation/testing	A manufactured floor assembly with supported attachment details

Western Train advertises custom parts, regauging and air brakes on passenger coaches. Severn Lamb offers custom rail configurations and documents a carriage/running-gear retrofit. These are credible inquiry

channels, but the public pages do not establish a bare-bogie rating or mounting pattern for this coach. Gauge matching is insufficient.

MGM's training manual distinguishes service and spring ports and model-specific mounting/release procedures. Require the exact selected model installation manual; farm crew must not open or modify a power-spring enclosure. Plascore offers engineered closeouts and inserts; this supports procuring finished panel interfaces rather than treating an unqualified skin as a fastener anchor. No component acceptance or supplier response is claimed.

Shop traveler and prototype tests

The full package contains **13 traveler steps** with named responsibility, required inputs, measured records and signoff gates, and **12 separate test records**. All acceptance and result fields remain blank. The critical sequence is:

Step group	Lead	Record and release condition
Scope and material	Vehicle engineer; shop inspector	Selected option IDs, source revisions, material heat/lot, wall/corner/plate readings
Empty fit-up and tight-end access	Farm crew with shop layout lead	Actual gaps, tool access, panel removal and collision photographs
Joint and fixture qualification	Welding engineer; test engineer	Accepted WPS/qualification, specimen/fixture drawing, calibration and abort limits
Coupon manufacture and inspection	Qualified shop; inspector	Fit-up, welder/joint trace, distortion, visual/NDE results, repairs before closure/coating
Seat/panel assembly and test	Supplier installers; test engineer	Measured stack, installed hardware, load/strain/displacement histories, unloading and residual set
Correlation and complete frame	Vehicle engineer; fabricator	Model/test reconciliation, full-frame mass/CG, support reactions, dimensional and weld records
Vehicle integration	Bogie/brake integrator	Separate empty/ballasted articulation, brake and running tests before commissioning

AWS D15.1/D15.1M:2025 is a directly relevant welding-basis candidate. Its public preview verifies the edition and scope, not project compliance. The responsible engineer must adopt the project basis and obtain the full requirements before selecting procedures, qualifications and inspection acceptance.

The test matrix covers empty access, unloaded dimensional survey, seat vertical, fore/aft, lateral/combined loads, CM6 local joint, thin outer seat-rail receiver, panel bearing/retention, bogie bolster assembly, fatigue, full frame and later running/brake integration. Every loaded test needs approved levels, combinations, staging, duration, fixture ratings, instrument ranges, pass limits, abort limits and unloading instructions before loading starts.

For traceability, the prior 305 kg rigid-bench/1g fore-aft sensitivity at 24 in CG gave 2.991028 kN foot compression,1.495514 kN uplift and 0.747757 kN shear. The earlier CM6 floor sensitivity gave 5.971 kN end shear and 1.618 kNm fixed-end moment. These are analytical reference demands, not proof-load or acceptance values. The full duty spectrum, stress ranges, cycle count and fatigue acceptance are still required; a successful static coupon cannot establish a service life or release the complete frame.

Farm crew can build empty nonstructural templates, record measurements, study handling/service access and prepare storage. Primary support weldments, loaded fixtures, examinations, brake stored-energy work and engineering release belong to the qualified specialists identified in the traveler. Physical fabrication release follows accepted evidence; another drawing revision alone does not constitute that evidence.

Primary source access date: **19 October 2026 UTC / 18 October Pacific**. Source JSON includes URLs, evidence limits and source-geometry hashes. No physical test or professional approval has been performed.

04 | Remaining development sequence

The remaining development sequence coordinates the subsystem work and the evidence needed for release. A completed desktop package is not permission to fabricate or operate.

Stage	Next coordinated package	Main external dependency before release
3J - Current	First-frame validation, load-path findings and first-article evidence plan	Accepted loads, bogie supports, local/fatigue design and qualified fabrication review
3K	Real air-brake components, trainline controls, failure behavior, torque/air/heat/timing budgets and bench test plan	Actuator/foundation matching, supplier data and validated full-consist response
3L	Bogies, wheels, axles, suspension, support footprint and wheel/rail compatibility	Actual rated undercarriage interfaces and vehicle-dynamics review
3M	Rated coupler/drawbar alternatives, frame attachments and protected inter-car services	Hardware ratings, fatigue/retention design and articulation evidence
3N	Locomotive propulsion, cooling, adhesion, gearing, controls and exhaust	Current Danfoss matching data, selected OEM ratings and accepted installation
3O-3Q	Coach body/interior; HVAC/electrical; accessible coach, boarding and evacuation	Real panel/seat/door/equipment interfaces and professional/accessibility review
3R	Track, drainage, station, storage and farm-crew installation interfaces	Survey, wheel/rail system, actual alignment/stop locations and approvals
3S	One coordinated configuration and controlled design review	Closed critical interfaces and actual signed release evidence
4	Procurement sequence, parts schedules, prototype fixtures and inspection traveler	Approved specifications, actual quotes and owner order authorization
5-6	Fabrication and track/site installation planning, traceability and as-built records	Released drawings/procedures, specialist oversight and actual inspection results
7	Instrumented commissioning, training, evacuation, maintenance and handover	Witnessed accepted tests and required engineering/authority/insurer release
8	Operating review, wear/reliability records and second-train readiness	Actual operating evidence and owner decision

Useful desktop work proceeds while external evidence is pending. Dependencies are explicit: for example, brake work informs bogie clearance and coupler loading, while the accepted bogie support interface resolves the structural release. Track typical sections can proceed before the North Pole location is fixed; final alignment cannot.

The program will maintain a costed procurement/construction schedule and alternatives as scope becomes supportable. Prices, budgets and business projections belong in owner-only records. The public site remains a technical review resource for Ralph, suppliers and fabricators.

No messages, orders or extra distribution are authorized by automatic continuation. Inquiry packages can be prepared for later instruction. Fabrication dimensions requiring missing evidence remain open; the program will not repeat speculative connection studies indefinitely or mark unperformed tests complete. When only external dependencies remain, it will consolidate those blockers instead of manufacturing another nominal phase.

The saved program state records outputs and dependencies. Phase 3K starts with dated MGM chamber and Bendix control research; road-trailer behavior does not establish a validated rail-train system.

05 | Review, formats and release limits

Deliverable	Use
Four dimensioned sheets	Integrated member map; bogie support interface; representative seat-floor mockup; inspection datums. PDF/SVG and editable DXFs.
J1a model	110 member solids, with 68 identified section changes. Browser rotation and named STEP exchange solids; no hidden accepted mounting hardware.
Calculation source	Editable inputs and Python analytic solutions, numerical outputs and verification records. These are not a validated whole-vehicle finite-element or fatigue model.
Prototype package	Supplier return fields, fabrication and inspection traveler, test plan gates and blank acceptance records. Blank fields indicate absent evidence, not permission to choose any value.
Program state	Next-stage scope and evidence dependencies, preserving prior issues so automatic continuation builds on the latest saved revision.

The numeric verification checks cover reaction and moment balance, selected beam benchmarks, file/geometry coordination and bounds. They do not establish strength, weld quality, crashworthiness, fatigue life, braking, stability, accessibility approval or jurisdiction. No supplier return, physical test or professional approval was obtained in this revision.

An independent bounded review passed 21 checks on the corrected mass ledger, main reactions and moments, bogie support balances, an off-center load benchmark, and the 68-of-110 member changes. The initial mass-scope mismatch was corrected before issue. This review does not replace the responsible engineer.

This managed session has no supported browser interaction tool. Static assets, JavaScript syntax, model data and document paths are checked; live pointer interaction, WebGL display and mobile behavior have not been exercised. The new model includes static views and dimensioned drawings as alternate reading paths.

Dimensions govern; do not scale PDF images. Each drawing and model README defines its units. STEP carries exchange solids rather than a native feature tree. Python and JSON preserve editable inputs, while source comments identify analytical idealizations.